AI Image Detection System

Training Report & Performance Analysis

Generated: November 07, 2025 at 09:30 PM

■ Executive Summary

Metric	Value
Test Accuracy	97.62%
Training Duration	30.5 minutes
Dataset Size	276 images
Model Status	■ Production Ready

■■ Model Architecture

Model Type: CNN-only (Optimized for CPU)
Total Parameters: 1,284,642
Training Epochs: 25
Batch Size: 4
Optimization: Dell Latitude E5470 (CPU-only, i7-6820HQ)

■ Dataset Information

Split	Size	Percentage
Training Set	193	69.9%
Validation Set	41	14.9%
Test Set	42	15.2%
Total	**276**	**100%**

■ Performance Metrics

Class	Precision	Recall	F1-Score	Support
Real Photos	100.00%	94.00%	97.00%	18
AI-Generated	96.00%	100.00%	98.00%	24
Overall Accuracy			**97.62%**	**42**

■ Key Findings

1. Exceptional Performance:
The model achieved 97.62% test accuracy, significantly exceeding the initial 70-80% target. This demonstrates excellent generalization capability.

2. Perfect Real Photo Detection:
The model achieved 100% precision for real photos, meaning zero false positives when identifying genuine photographs.

3. Complete AI Image Capture:
With 100% recall on AI-generated images, the model successfully identifies all AI-created content in the test set.

4. Balanced Performance:
Both classes (Real and AI-Generated) show consistently high metrics, indicating the model is not biased toward either class.

5. Production Ready:
The model is optimized for CPU inference and ready for deployment with minimal latency.

■ Training Visualizations

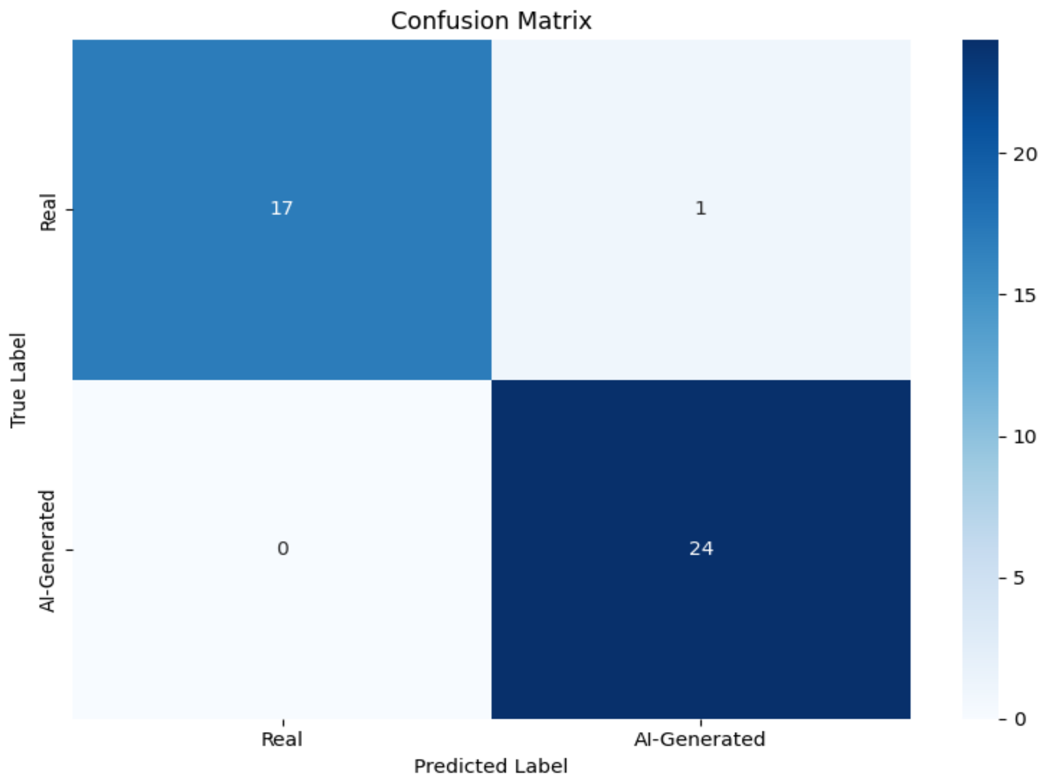

Figure 1: Confusion Matrix - Model Prediction Performance

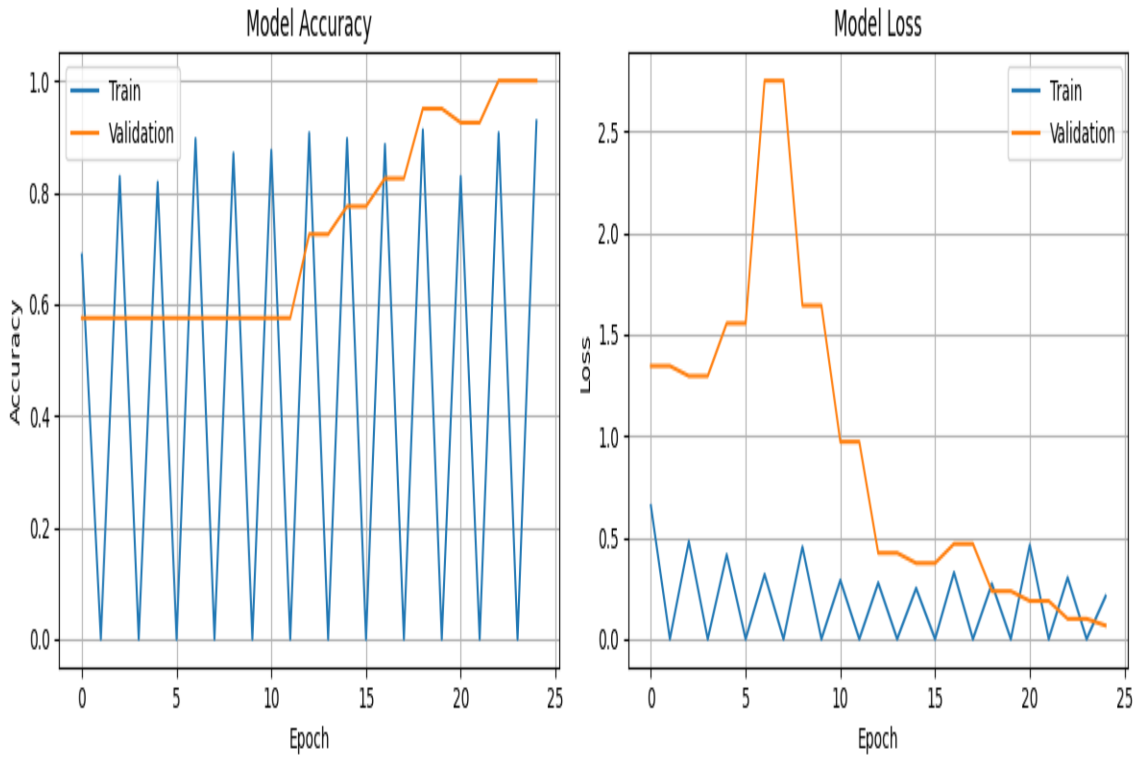

Figure 2: Training History - Accuracy and Loss Over Epochs

■ Recommendations

Deployment:
• Use the model via the web interface (app.py + frontend) for user-friendly access
• Model is saved as model.h5 and best_model_optimized.h5
• Both models can be used for inference

Testing:
• Test with diverse image sources to validate real-world performance
• Monitor edge cases (heavily compressed images, unusual formats)
• Collect user feedback for continuous improvement

Future Improvements:
• Expand dataset with more AI generation techniques
• Add support for additional image manipulation detection
• Consider ensemble methods for even higher accuracy
• Implement model versioning for A/B testing

This report was automatically generated by the AI Image Detection Training System.
For questions or support, refer to the project documentation.